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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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Logbook 7

1.1 Aim

The aim of this lab is to learn some fundamental Linux command and create a script file for recording the terminal session.

1.2 Objectives

The logbook assignment is given to year 2 students for Network Operating System module. This is the week 7 logbook assignment where students are required to learn some basic Linux command. It was done at the workshop of the module.

The objectives of this logbook are listed below:

- To gain knowledge on basic command of Linux.
- To acknowledge the significance of script when documenting the steps.
- To learn and understand the difference between different file and directories listing mode.
- To learn the command to install the utility packages in Linux.
- To get the knowledge on how to create single line and multiline file.

1.3 Required tools and concepts

The tools and concepts required for are:

Tools

- Debian Linux operating system and its terminal to run the command.
- Script command to record every step for documentation.
- Some basic command of Linux to run on Debian terminal.
- Finger utility package installed on Linux to see system details.

Concepts

- Different file listing mode concept to view files in different level.
- Creation of file with echo and cat.
- Recording of script for command documentation.
- Basic of file manipulation like combining two files together.

1.4 Steps required for lab

First, open Debian terminal in your host OS to run command of Linux and follow these steps:

Record every step done in terminal inside a1script file

In the Debian terminal, to record every step you do write the command "script filename" and press enter.

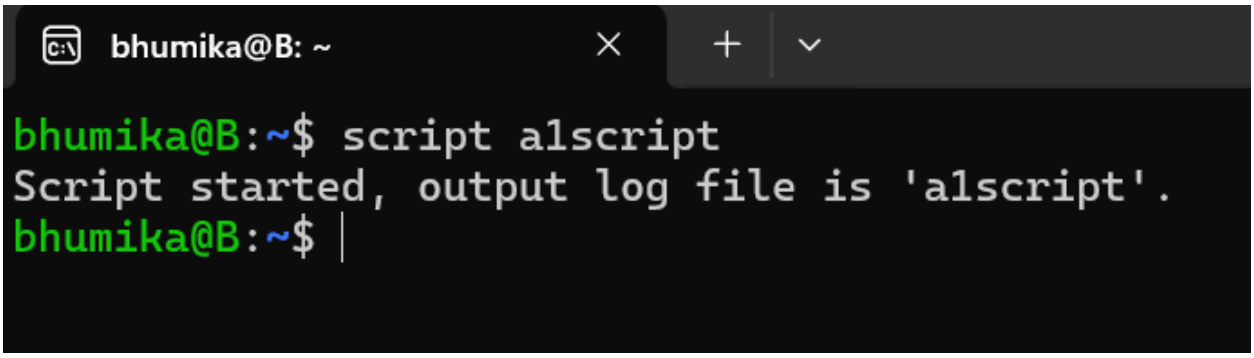

Figure 1: Starting Script

Displays the username

To print the username of the present Debian user, write the command "whoami" and it will show you your username.

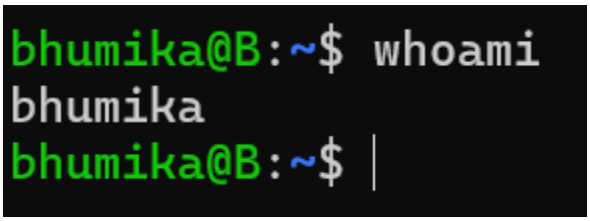

Figure 2: Displaying the present user's username

Displays all the user present in the Debian Linux system

To print the username of all the user on Debian Linux system, write the command "who" and it will show you the username of all the user.

```
bhumika@B:~$ who
bhumika  pts/1        2025-12-07 03:05
bhumika@B:~$
```

Figure 3: Displaying all the user in the Debian Linux

Display the information about the system

To display the details of the system, write the command "finger username" and click enter. We have to make sure that the finger utility package is installed before running the command. To download the finger utility package, we have to write the command "sudo install apt finger".

```
bhumika@B:~$ sudo apt install finger
[sudo] password for bhumika:
Installing:
  finger

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 19.8 kB
  Space needed: 50.2 kB / 1,026 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 finger amd64 0.17-17 [19.8 kB]
Fetched 19.8 kB in 0s (57.6 kB/s)
Selecting previously unselected package finger.
(Reading database ... 8706 files and directories currently installed.)
Preparing to unpack .../finger_0.17-17_amd64.deb ...
Unpacking finger (0.17-17) ...
Setting up finger (0.17-17) ...
```

Figure 4: Downloading finger utility package

Since the finger utility package was not installed before after installing the utility package now, we have to run the command "finger username".

```
bhumika@B:~$ finger bhumika
Login: bhumika                          Name:
Directory: /home/bhumika                Shell: /bin/bash
Never logged in.
No mail.
No Plan.
bhumika@B:~$
```

Figure 5: Displaying the information of the system

Display the current date

To print today's date and current time, write the command "date" and press enter.

```
bhumika@B:~$ date
Sun Dec  7 03:14:52 AM UTC 2025
bhumika@B:~$
```

Figure 6: Displaying current date and time

File listing in different mode

File can be list in different mode such as default mode, all mode (which includes hidden files too), and detailed mode.

Listing visible files

The command "ls" lists the non-hidden normal files.

```
bhumika@B:~$ ls
a1script
bhumika@B:~$
```

Figure 7: file listing command

Listing files including hidden ones

The command "ls" does not list hidden files. To list the hidden ones as well we have to write the command "ls -a".

```
bhumika@B:~$ ls -a
.  ..  a1script  .bash_history  .bash_logout  .bashrc  .profile  .sudo_as_admin_successful
bhumika@B:~$
```

Figure 8: Including hidden files

Listing files including hidden ones with other details

To list the files details such as file size, permissions, timestamp, file name and so on. We have to write the command "ls -a -l" and click enter.

```
bhumika@B:~$ ls -a -l
total 28
drwx------ 2 bhumika bhumika 4096 Dec  7 03:07 .
drwxr-xr-x 3 root    root    4096 Dec  4 07:15 ..
-rw-r--r-- 1 bhumika bhumika 4096 Dec  7 03:12 a1script
-rw------- 1 bhumika bhumika   18 Dec  4 07:15 .bash_history
-rw-r--r-- 1 bhumika bhumika  220 Dec  4 07:15 .bash_logout
-rw-r--r-- 1 bhumika bhumika 3526 Dec  4 07:15 .bashrc
-rw-r--r-- 1 bhumika bhumika  807 Dec  4 07:15 .profile
-rw-r--r-- 1 bhumika bhumika    0 Dec  4 07:15 .sudo_as_admin_successful
bhumika@B:~$
```

Figure 9: Listing file details

Displaying what is inside the file

To know what the file contains, we have to write the command "cat /etc/passwd" and click enter.

```
bhumika@B:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:996:996:System Message Bus:/nonexistent:/usr/sbin/nologin
bhumika:x:1000:1000::/home/bhumika:/bin/bash
bhumika@B:~$
```

Figure 10: displaying what's in a file

Creating single line input file

To create a single line file, we have to write the command "echo "This is a one-line file" > test1".

```
bhumika@B:~$ echo "This is a one-line file" > test1
bhumika@B:~$
```

Figure 11: Creating the single line file

Creating multi line input file

To create a multiline file, we have to write the command "cat > test2
This is second file.
created with cat." and click CTRL + D to exit from the file input.

```
bhumika@B:~$ cat > test2
This is second file
created with cat.bhumika@B:~$
```

Figure 12: Creating the multiline file

Showing the recently created file exists

To show the file made we have to write the command "cat filename".
We have created two files. For accessing the first file we have to write the command "cat test1".

```
bhumika@B:~$ cat test1
This is a one-line file
bhumika@B:~$
```

Figure 13: Displaying the single line file

Now, for accessing the second file we have to write the command "cat test2".

```
bhumika@B:~$ cat > test2
This is second file
created with cat.bhumika@B:~$
```

Figure 14: Displaying the multiline file

Combining the two files

To combine the two files together, we have to write the command "cat file1name file2name".

```
bhumika@B:~$ cat test1 test2
This is a one-line file
This is second file
created with cat.bhumika@B:~$
```

Figure 15: combining the two files together

Exiting the script

To exit from the a1script we simply have to write the command "exit" and the recording will stop.

```
bhumika@B:~$ exit
exit
Script done.
bhumika@B:~$
```

Figure 16: exiting the script

1.5 Conclusion

This aim of this lab was to learn the fundamental Linux commands and create a script file to log the operations. It included running some basic commands such as whoami, who, finger, date, ls and so on.

The hands-on practice helped to understand the process of creating a one-line and multiple-lines files with the help of echo and cat, respectively, and to view what is in the file and merge files. The script command recorded all the command and output to get documentation of the entire session.

From this lab, multiple knowledge was acquired. Some of them are navigating in Linux terminal, creating files, and performing some basic manipulations. It showed the actual command line working process that is relevant in day-to-day system administration activities.

Overall, this workshop had set a good foundation in Linux skills which are very important in the management of network operating systems and server environment.

1.6 Appendix

```
cat a1script
Script started on 2025-12-07 03:07:49+00:00 [TERM="xterm-256color"
TTY="/dev/pts/0" COLUMNS="156" LINES="41"]
bhumika@B:~$ whoami
bhumika
bhumika@B:~$ who
bhumika   pts/1        2025-12-07 03:05
bhumika@B:~$ finger bhumika
finger: command not found
bhumika@B:~$ sudo apt install finger
[sudo] password for bhumika:
Installing:
  finger

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 19.8 kB
  Space needed: 50.2 kB / 1,026 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 finger amd64 0.17-17
[19.8 kB]
Fetched 19.8 kB in 0s (57.6 kB/s)
Selecting previously unselected package finger.
(Reading database ... 8706 files and directories currently installed.)
Preparing to unpack .../finger_0.17-17_amd64.deb ...
Unpacking finger (0.17-17) ...
Setting up finger (0.17-17) ...
bhumika@B:~$ finger bhumika
Login: bhumika                          Name:
Directory: /home/bhumika                Shell: /bin/bash
Never logged in.
No mail.
No Plan.
bhumika@B:~$ date
Sun Dec  7 03:14:52 UTC 2025
bhumika@B:~$ ls
a1script
bhumika@B:~$ ls -a
.  ..  a1script  .bash_history  .bash_logout  .bashrc  .profile
.sudo_as_admin_successful
bhumika@B:~$ ls-a
ls-a: command not found
bhumika@B:~$ ls -a -l
total 28
drwx------ 2 bhumika bhumika  4096 Dec  7 03:07 .
drwxr-xr-x 3 root    root     4096 Dec  4 07:15 ..
-rw-r--r-- 1 bhumika bhumika  4096 Dec  7 03:12 a1script
-rw------- 1 bhumika bhumika    18 Dec  4 07:15 .bash_history
-rw-r--r-- 1 bhumika bhumika   220 Dec  4 07:15 .bash_logout
-rw-r--r-- 1 bhumika bhumika     0 Dec  4 07:15 .bashrc
-rw-r--r-- 1 bhumika bhumika   807 Dec  4 07:15 .profile
-rw-r--r-- 1 bhumika bhumika     0 Dec  4 07:15 .sudo_as_admin_successful
bhumika@B:~$ cat /etc/passwd
```

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time
Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
bhumika:x:1000:1000:,,,:/home/bhumika:/bin/bash
bhumika@B:~$ echo "This is a one-line file" > test1
bhumika@B:~$ cat > test2
This is second file
created with cat.bhumika@B:~$
bhumika@B:~$ cat test1
This is a one-line file
bhumika@B:~$ cat test2
This is second file
created with cat.bhumika@B:~$
bhumika@B:~$ cat test1 test2
This is a one-line file
This is second file
created with cat.bhumika@B:~$
bhumika@B:~$ exit
exit

Script done on 2025-12-07 03:23:03+00:00 [COMMAND_EXIT_CODE="0"]
```